

Superstore Profit Analysis Using Power BI

Abstract

This project uses Power BI to analyze Superstore sales and profit data to identify business performance across regions, categories, segments, sub-categories, and time.

The objective is to transform raw retail transaction data into interactive dashboards that help management monitor profit trends, identify profitable product categories, compare regional performance, and support strategic business decisions through data visualization.

Problem Statement

Retail businesses generate large amounts of transactional data every day. Without proper analysis, it becomes difficult to identify profitable products, high-performing regions, and loss-making areas.

This project aims to answer:

- Which region generates the highest profit?
- Which product category contributes the most profit?
- Which customer segment is the most profitable?
- Which sub-categories generate losses?
- How does profit change over time?
- Which states contribute most to business performance?

Dataset Information

Dataset Name: Superstore Sales Dataset

Data Source: Sample Superstore Dataset

Type of Data: Secondary Data

Dataset Attributes

Order ID, Order Date, Ship Date, Customer ID, Customer Name, Segment, Country, State, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit

Tools Used

- Microsoft Power BI
- Microsoft Excel (Data Cleaning)
- DAX
- Power Query

System Flow Diagram



Dashboard Components

KPI Cards: Total Sales (62.26K), Total Profit (4.40K), Total Quantity (1K).

Visuals: Profit by Year & Month, Region, Category, Segment, Sub-Category, Profit Matrix, State Filter.

Analysis Performed

- 1) Total Sales: 62.26K
- 2) Total Profit: 4.40K
- 3) Total Quantity: 1K
- 4) Profit by Region: South performs best.
- 5) Profit by Category: Technology contributes the highest profit.
- 6) Profit by Segment: Corporate is more profitable than Consumer.

- 7) Profit Trend: Monthly analysis from 2016–2019 shows seasonal fluctuations.
- 8) Profit by Sub-Category: Phones lead profits; Tables and Binders incur losses.
- 9) Regional comparison identifies strongest markets.
- 10) State slicer enables detailed analysis.

Business Insights

- Technology is the strongest profit-generating category.
- South region delivers the highest profitability.
- Corporate customers contribute greater profits.
- Phones are the most profitable sub-category.
- Tables and Binders reduce overall profitability.
- Profit trends fluctuate over time.

Key Findings

- Technology is the most profitable category.
- Phones generate the highest profit.
- Tables and Binders are loss-making.
- South region performs best.
- Corporate customers generate higher profits.
- Interactive filters support state-level analysis.

Recommendations

- 1) Promote high-profit products such as Phones and Technology.
- 2) Review pricing and discounts for Tables and Binders.
- 3) Expand marketing in high-performing regions.
- 4) Strengthen Corporate customer relationships.
- 5) Monitor monthly profit trends.
- 6) Use state-level insights for regional planning.

Learning Outcomes

Power BI Data Import, Power Query, Data Cleaning, Data Modeling, Relationships, DAX Measures, KPI Cards, Slicers, Charts, Matrix Visuals, Dashboard Design, Interactive Filtering.

Conclusion

This Power BI dashboard transforms retail sales data into meaningful business insights. It highlights profitable regions, customer segments, and product categories while identifying areas requiring improvement, enabling informed business decisions.